

2. Literature Analysis Summary

2.1 Backbone Improvements

2.1.1 Role of the Backbone

The backbone serves as the feature extraction component of an object detection network. Its primary responsibility is to learn meaningful representations from the input image before passing them to subsequent stages of the detector. In road damage detection, the backbone must effectively capture features corresponding to cracks, potholes, and other surface defects that often vary significantly in size, texture, and orientation. Poor feature extraction at this stage negatively affects every downstream component of the network.

2.1.2 Backbone Modules Reviewed

The literature review identified several approaches for improving the YOLOv8 backbone. The reviewed modules included GhostConv, GSConv, DAT, BoT, SOM, ODConv, BoTNet, SBS, and EMA_Faster_C2f.

Although these modules pursue the same objective, they employ different strategies. GhostConv, GSConv, and SBS mainly focus on lightweight convolution to reduce parameters and computational cost. ODConv and BoTNet introduce more adaptive feature learning through dynamic convolution and transformer concepts. DAT utilizes deformable attention to improve spatial feature extraction, whereas EMA_Faster_C2f combines Faster Blocks, Partial Convolution, and EMA attention into a lightweight feature extraction module.

2.1.3 Comparative Analysis

The reviewed literature shows that lightweight convolution methods provide good computational efficiency but generally produce smaller improvements in detection accuracy. Dynamic convolution methods improve adaptability but increase implementation complexity and computational cost.

Transformer-based approaches consistently demonstrate stronger feature learning, particularly for irregular and elongated road damages. Among all reviewed methods, DAT showed the strongest experimental evidence. By applying deformable attention, it enables the network to concentrate on informative image regions while maintaining computational efficiency. The reported ablation studies also demonstrated one of the largest individual performance improvements among all reviewed backbone modules.

EMA_Faster_C2f also demonstrated promising results by combining multiple lightweight improvements into a single module. However, compared with DAT, it has been evaluated in fewer studies, providing less experimental evidence regarding its general applicability.

2.1.4 Conclusion

Considering experimental performance, implementation feasibility, literature support, and compatibility with YOLOv8, DAT was selected as the backbone enhancement for the proposed architecture.

EMA_Faster_C2f demonstrated promising results and remains a strong candidate for future experimentation. However, DAT provides stronger experimental validation and better aligns with the design philosophy of improving feature extraction while maintaining architectural simplicity.

2.2 Neck Improvements

2.2.1 Role of the Neck

The neck combines feature maps generated by the backbone before they are forwarded to the detection head. Since road damages appear at multiple scales, effective feature fusion is essential for accurately detecting both small cracks and large potholes. A well-designed neck improves the interaction between low-level spatial information and high-level semantic features, enabling the detector to build richer multi-scale representations.

2.2.2 Neck Modules Reviewed

The reviewed literature proposed several improvements to the standard YOLOv8 PAN-FPN architecture. The primary feature fusion modules included VoVGSCSP, MAF, and AE-FPN.

VoVGSCSP focused on improving computational efficiency while maintaining effective information flow. MAF introduced adaptive feature weighting to improve multi-scale feature aggregation. AE-FPN further enhanced feature fusion by integrating adaptive feature aggregation with EMA attention, allowing stronger interaction between feature maps from different network levels.

2.2.3 Comparative Analysis

One of the strongest observations from the literature review was that improvements to feature fusion consistently produced significant performance gains across different models. This indicates that effective communication between feature maps is critical for detecting road damages with varying sizes and shapes.

VoVGSCSP provided computational efficiency with moderate improvements in detection accuracy. MAF demonstrated effective adaptive feature weighting and produced strong

ablation results. AE-FPN combined adaptive aggregation with attention mechanisms to strengthen feature interaction across multiple scales. Compared to the other reviewed neck designs, AE-FPN consistently demonstrated larger performance improvements while remaining relatively straightforward to integrate into the YOLO architecture.

2.2.4 Conclusion

AE-FPN was selected as the neck module because it demonstrated the strongest balance between feature fusion capability, experimental evidence, and implementation feasibility.

MAF remains a suitable alternative for future investigation, but AE-FPN better satisfies the objectives of the proposed hybrid architecture.

2.3 Detection Head Improvements

2.3.1 Role of the Detection Head

The detection head is responsible for performing object classification and bounding box regression using the feature maps produced by the backbone and neck. The overall detection accuracy depends not only on the quality of extracted features but also on how effectively the detection head converts these features into final predictions. An efficient detection head should improve localization accuracy while minimizing computational overhead.

2.3.2 Detection Head Modules Reviewed

Several detection head designs were reviewed during the literature survey, including the original YOLOv8 detection head, Efficient Head, SEAM, MultiSEAM, WTC, and LSCD-Head.

Each module proposed different improvements to the prediction stage. Some focused on lightweight computation, while others introduced improved feature refinement or frequency-domain processing. LSCD-Head introduced shared convolution layers together with Group Normalization and scale adjustment mechanisms to improve prediction efficiency.

2.3.3 Comparative Analysis

The original YOLOv8 detection head provides a reliable baseline but still contains redundant computations that can be optimized.

Efficient Head, SEAM, and MultiSEAM demonstrated measurable improvements over the baseline architecture but were evaluated in relatively few studies. WTC introduced an

innovative frequency-domain prediction mechanism and achieved promising results. However, the available experimental evidence was limited compared to other detection head designs.

LSCD-Head consistently demonstrated strong performance across multiple independent studies. By utilizing shared convolutions and Group Normalization, it reduces computational redundancy while improving prediction quality across multiple feature scales. The repeated validation of LSCD-Head across different architectures increases confidence in its effectiveness.

2.3.4 Conclusion

LSCD-Head was selected as the detection head for the proposed architecture because it demonstrated consistent improvements across multiple studies while maintaining computational efficiency.

Although WTC remains an interesting alternative due to its novel design, LSCD-Head provides stronger experimental support and better complements the selected backbone and neck modules.

2.4 Loss Function Improvements

2.4.1 Role of the Loss Function

The loss function guides the optimization process during training by measuring the difference between predicted and ground truth bounding boxes. Although the backbone, neck, and detection head determine how features are extracted and predictions are generated, the loss function influences how efficiently the network learns from its errors. An improved localization loss can enhance bounding box accuracy, particularly for irregular road damages such as cracks and potholes.

2.4.2 Loss Functions Reviewed

The standard YOLOv8 model employs Complete Intersection over Union (CIoU) loss together with Distribution Focal Loss (DFL) and classification loss during training. Several alternative localization losses have been proposed in recent years, including Wise-IoU (WIoU), Focal-EIoU, and MPDIoU.

Among the reviewed literature, MPDIoU was the only loss function evaluated through a dedicated ablation study within a road damage detection framework.

2.4.3 Comparative Analysis

Wise-IoU and Focal-EIoU have demonstrated promising localization performance in general object detection research. However, they were not directly evaluated within the selected road damage detection papers, limiting the ability to compare their effectiveness under identical experimental conditions.

MPDIoU improves localization by considering the distance between corresponding corner points in addition to the overlap between predicted and ground truth bounding boxes. This enables more accurate localization when bounding boxes overlap but remain slightly misaligned. The reviewed ablation study demonstrated consistent improvements in localization accuracy while introducing minimal computational overhead.

Another important observation is that replacing the localization loss only affects the training stage and does not increase inference complexity. This makes MPDIoU an efficient enhancement that can be combined with other architectural improvements without affecting deployment performance.

2.4.4 Conclusion

MPDIoU was selected as the localization loss for the proposed architecture because it demonstrated improved bounding box regression through dedicated experimental evaluation while requiring minimal implementation effort.

Wise-IoU and Focal-EIoU remain promising alternatives for future experimentation, but the currently available evidence was not sufficient to justify their inclusion in the proposed architecture.

2.5 Chapter Summary

The literature review demonstrates that recent improvements in YOLO-based road damage detection are no longer limited to a single component of the network. Instead, researchers have focused on enhancing multiple stages of the detection pipeline, including feature extraction, feature fusion, prediction, and localization. Each reviewed module addressed a specific limitation of the baseline architecture and contributed differently to overall detection performance.

A key observation from the reviewed studies is that feature fusion and attention-based mechanisms consistently produced larger performance improvements than isolated lightweight modifications. Furthermore, modules supported by comprehensive ablation studies and repeated validation across multiple papers provided stronger evidence for practical adoption.

Based on the comparative analysis presented in this chapter, one module was selected from each major component of the detection pipeline. **DAT** was chosen to enhance feature

extraction within the backbone, **AE-FPN** was selected to improve multi-scale feature fusion, **LSCD-Head** was adopted to strengthen the prediction stage, and **MPDIoU** was selected to improve localization during training.

These selections form the foundation of the proposed hybrid architecture presented in the next chapter. Rather than combining multiple modules with overlapping functionality, the proposed design integrates complementary improvements at different stages of the YOLOv8 framework. This modular approach aims to improve detection accuracy while maintaining computational efficiency and simplifying the evaluation of each architectural enhancement.

3. Proposed Hybrid Architecture Design

3.1 Design Objectives

The primary objective of the proposed architecture is to improve road damage detection by combining the most effective architectural improvements identified during the literature review. Rather than developing an entirely new object detection framework, this work builds upon the proven YOLOv8 architecture and selectively integrates modules that demonstrated consistent improvements in feature extraction, feature fusion, prediction, and localization.

The proposed design follows four primary objectives:

- Improve feature extraction for complex and irregular road damages.
- Strengthen multi-scale feature fusion to improve the detection of both small and large defects.
- Enhance prediction efficiency without significantly increasing computational complexity.
- Improve bounding box localization while maintaining real-time inference capability.

Another important design objective is modularity. Each selected module addresses a different stage of the detection pipeline, allowing every modification to be evaluated independently through ablation studies. This makes the architecture easier to analyze, optimize, and extend in future work.

3.2 Overall Architecture Overview

The proposed model adopts **YOLOv8m** as the baseline architecture. Among the baseline models evaluated during the initial experiments, YOLOv8m achieved the best balance between detection accuracy and computational efficiency. Instead of replacing the complete architecture, only the components that consistently demonstrated improvements in the reviewed literature are modified.

The proposed architecture consists of four primary improvements:

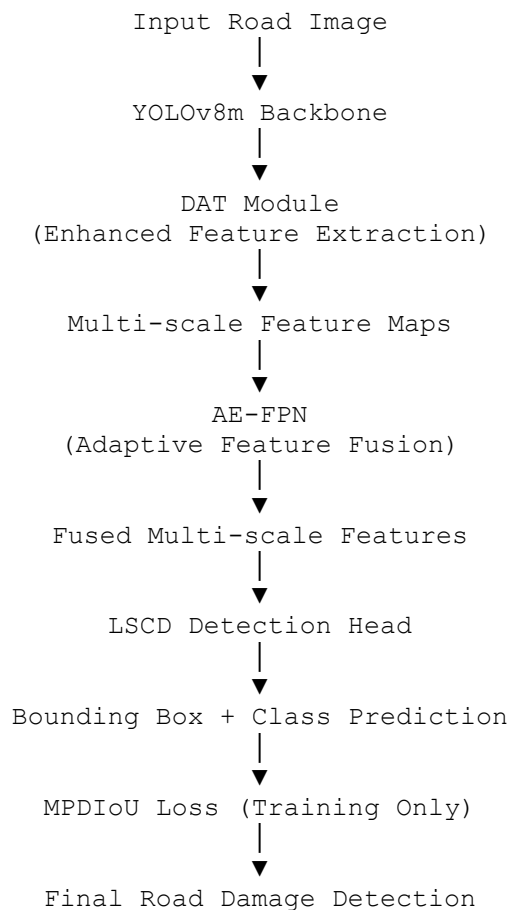
Component	Selected Module	Purpose
Backbone	DAT	Improve feature extraction
Neck	AE-FPN	Improve multi-scale feature fusion
Detection Head	LSCD-Head	Improve prediction efficiency
Loss Function	MPDIoU	Improve localization accuracy

These modifications target different stages of the network and therefore complement one another instead of introducing redundant functionality. The resulting architecture preserves the overall structure of YOLOv8 while incorporating recent advancements identified through the literature review.

3.3 Overall Detection Pipeline

The proposed detection framework follows the same general workflow as YOLOv8 while replacing selected components with improved alternatives.

The complete pipeline can be summarized as follows:



During inference, the architecture predicts road damages directly from the optimized detection head. During training, MPDIoU replaces the default localization loss to improve bounding box regression without affecting inference speed.

3.4 Module Selection Strategy

Instead of integrating every improvement proposed in the literature, the proposed architecture follows a selective integration strategy. Multiple modules addressing the same architectural component were intentionally avoided because they often solve similar problems and may introduce unnecessary complexity or compatibility issues.

For each major component of YOLOv8, the literature was analyzed to identify the module with the strongest combination of:

- Experimental performance
- Literature support
- Ablation evidence
- Ease of implementation
- Compatibility with YOLOv8

Following this process, only one module was selected from each architectural category. This strategy simplifies implementation, reduces redundancy, and makes it easier to measure the contribution of each module during experimental evaluation.

3.5 Rationale Behind Module Selection

The proposed architecture was designed based on the principle that different stages of an object detector solve different problems. Therefore, each selected module addresses a unique limitation of the baseline YOLOv8 architecture.

DAT enhances the backbone by improving feature extraction through deformable attention. This enables the network to better capture irregular road damage patterns and long-range spatial relationships.

AE-FPN improves the neck by enhancing feature fusion across multiple scales. This helps preserve both low-level spatial information and high-level semantic information, resulting in stronger representations for road damages of different sizes.

LSCD-Head strengthens the prediction stage by reducing redundant computations while improving localization and classification performance through shared convolutions and improved normalization.

MPDIoU enhances the training process by improving bounding box regression. Since it only modifies the optimization objective during training, it provides better localization accuracy without increasing inference complexity.

Together, these modules improve different stages of the detection pipeline without overlapping responsibilities, making the overall architecture both efficient and modular.

3.6 Expected Advantages of the Proposed Architecture

Based on the reviewed literature and the selected module combination, the proposed architecture is expected to provide several advantages over the baseline YOLOv8 model.

Expected improvements include:

- Better feature extraction for irregular cracks and potholes.
- Improved multi-scale feature representation.
- Stronger detection of small road damages.
- More accurate bounding box localization.
- Reduced redundant computations in the prediction stage.
- Better balance between detection accuracy and computational efficiency.

Because each selected module has demonstrated positive experimental results individually, integrating them into a unified architecture is expected to produce cumulative improvements across multiple evaluation metrics, including Precision, Recall, mAP@50, and mAP@50-95.

3.7 Planned Experimental Evaluation

The effectiveness of the proposed architecture will be evaluated through a series of controlled experiments using the RDD2022 dataset.

The evaluation will include:

Baseline Comparison

- YOLOv8n
- YOLOv8s
- YOLOv8m
- Proposed Hybrid YOLO

Ablation Study

Individual contribution of:

- DAT
 - AE-FPN
 - LSCD-Head
 - MPDIoU
-

Dataset Evaluation

- Complete RDD2022 dataset
 - India subset evaluation
-

SAHI Evaluation

- Proposed Hybrid YOLO
 - Proposed Hybrid YOLO + SAHI
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Evaluation Metrics

- Precision
 - Recall
 - mAP@50
 - mAP@50-95
 - Number of detections
 - Inference speed
 - Model complexity
-

3.8 Chapter Summary

This chapter presented the overall design of the proposed hybrid architecture developed from the findings of the literature review. Instead of modifying every component of the YOLOv8 network, the architecture selectively integrates one validated improvement into each major stage of the detection pipeline. DAT enhances feature extraction, AE-FPN improves feature fusion, LSCD-Head strengthens the prediction stage, and MPDIoU improves localization during training.

The proposed architecture maintains the overall structure of YOLOv8 while incorporating complementary improvements supported by previous research. This modular design simplifies implementation and enables systematic evaluation through ablation studies and benchmark comparisons. The following chapter will examine each selected module in greater detail, explaining its working principles, architectural integration, and expected contribution to the overall detection performance.

4. Detailed Module Design

4.1 DAT (Deformable Attention Transformer)

4.1.1 Purpose

DAT was selected to improve the feature extraction capability of the YOLOv8 backbone. Traditional convolution operations apply fixed receptive fields, making it difficult to capture irregular road damages such as thin cracks or elongated potholes. DAT overcomes this limitation by introducing deformable attention, allowing the network to focus on the most informative image regions rather than processing every spatial location equally.

4.1.2 Why DAT Was Selected

Several backbone improvements were reviewed during the literature survey, including GhostConv, GSConv, SBS, ODConv, BoTNet, and EMA_Faster_C2f.

DAT was selected because it demonstrated:

- Strong experimental improvements
 - Large gains in ablation studies
 - Better feature extraction
 - Good compatibility with YOLOv8
 - Balanced computational complexity
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4.1.3 Expected Contribution

Expected improvements include:

- Better representation of long cracks
 - Better contextual understanding
 - Stronger small-object detection
 - Improved feature quality before feature fusion
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4.1.4 Planned Integration

DAT will replace selected C2f feature extraction blocks inside the YOLOv8m backbone while preserving the overall network structure.

4.2 AE-FPN (Adaptive Enhanced Feature Pyramid Network)

4.2.1 Purpose

AE-FPN improves feature fusion by combining features from multiple scales more effectively than the standard PAN-FPN architecture. Since road damages vary significantly in size, stronger feature interaction improves detection robustness.

4.2.2 Why AE-FPN Was Selected

Compared with VoVGSCSP and MAF,

AE-FPN demonstrated:

- strongest feature fusion
 - best ablation improvements
 - effective attention mechanism
 - relatively simple integration
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4.2.3 Expected Contribution

Expected improvements include:

- Better multi-scale learning
 - Stronger small crack detection
 - Improved preservation of spatial information
 - Better feature communication
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4.2.4 Planned Integration

The standard YOLOv8 PAN-FPN neck will be replaced with AE-FPN while preserving feature map resolutions and maintaining compatibility with the detection head.

4.3 LSCD Detection Head

4.3.1 Purpose

LSCD-Head improves the final prediction stage by reducing redundant computations while improving localization and classification efficiency.

4.3.2 Why LSCD Was Selected

Compared with

- Efficient Head
- SEAM
- MultiSEAM
- WTC

LSCD demonstrated

- strongest experimental evidence
 - repeated validation
 - lightweight prediction
 - easier integration
-

4.3.3 Expected Contribution

Expected improvements include

- Faster prediction
 - Better localization
 - Better multi-scale prediction
 - Lower parameter redundancy
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4.3.4 Planned Integration

The original YOLOv8 detection head will be replaced with LSCD while maintaining the same prediction outputs.

4.4 MPDIoU Loss

4.4.1 Purpose

MPDIoU improves bounding box regression by considering both overlap and corner point distance during optimization.

4.4.2 Why MPDIoU Was Selected

MPDIoU was the only localization loss directly evaluated through an ablation study within the reviewed literature.

Advantages include

- Better localization

- Minimal implementation effort
 - No inference overhead
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4.4.3 Expected Contribution

Expected improvements include

- More accurate bounding boxes
 - Better localization of elongated damages
 - Stable convergence
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4.4.4 Planned Integration

MPDIoU will replace the default CIoU localization loss during training while leaving inference unchanged.

4.5 Module Compatibility

One important objective during the architecture design process was ensuring that all selected modules improve different stages of the network without overlapping functionality.

Module	Network Stage	Primary Function
DAT	Backbone	Feature Extraction
AE-FPN	Neck	Feature Fusion
LSCD	Detection Head	Prediction
MPDIoU	Training	Localization

Because each module targets a different component, the proposed architecture is expected to benefit from complementary improvements rather than redundant modifications.

4.6 Chapter Summary

This chapter presented the individual components selected for the proposed hybrid architecture. Each module was chosen based on its experimental evidence, implementation feasibility, and compatibility with YOLOv8. Together, these modules form the core of the proposed detection framework and provide a clear roadmap for implementation.

5. Integration Strategy

5.1 Integration Order

To minimize implementation complexity and simplify debugging, the modules will be integrated sequentially rather than simultaneously.

Planned implementation order:

1. Baseline YOLOv8m
2. DAT Integration
3. AE-FPN Integration
4. LSCD Integration
5. MPDIoU Integration
6. SAHI Integration

5.2 Compatibility Considerations

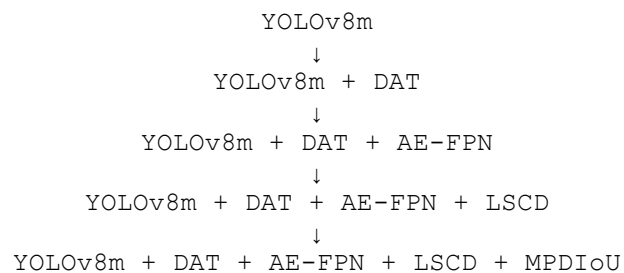
Before integrating each module, several compatibility factors must be verified:

- Tensor dimensions
- Feature map resolutions
- Channel consistency
- Weight loading compatibility
- Training stability

5.3 Planned Ablation Strategy

Each module will first be evaluated individually before constructing the final architecture.

Example sequence:



This sequential approach makes it possible to quantify the contribution of each architectural modification.

5.4 Expected Challenges

Potential implementation challenges include:

- Module compatibility
 - Pretrained weight adaptation
 - Increased training time
 - Hyperparameter tuning
 - Maintaining computational efficiency
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5.5 Chapter Summary

This chapter outlined the planned implementation strategy for integrating the selected modules into YOLOv8m. A sequential integration approach will simplify debugging, support systematic ablation studies, and ensure that each architectural modification can be independently evaluated.

6. Experimental Plan

6.1 Dataset

Primary Dataset - RDD2022

Additional Evaluation - India subset

6.2 Baseline Models

Comparison models include

- YOLOv8n
 - YOLOv8s
 - YOLOv8m
 - Proposed Hybrid YOLO
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6.3 Training Configuration

Training parameters to be finalized include:

- Image size
 - Batch size
 - Epochs
 - Learning rate
 - Optimizer
 - Data augmentation
 - Hardware configuration
-

6.4 Evaluation Metrics

Performance will be evaluated using

- Precision
 - Recall
 - mAP@50
 - mAP@50-95
 - F1-score
 - Number of detections
 - FPS (if measured)
 - Inference time
 - Model size
-

6.5 Planned Experiments

The proposed evaluation includes:

Experiment 1

Baseline comparison

Experiment 2

Module ablation study

Experiment 3

Complete hybrid architecture

Experiment 4

Hybrid architecture + SAHI

Experiment 5

India subset evaluation

6.6 Expected Outcomes

The proposed architecture is expected to:

- Improve feature extraction
 - Improve feature fusion
 - Improve localization
 - Increase detection accuracy
 - Maintain practical computational efficiency
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6.7 Future Extensions

Potential future work includes:

- YOLOv11 implementation
 - Domain adaptation
 - Real-time deployment
 - Mobile optimization
 - Crack segmentation integration
 - Multitask learning
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Final Remarks

This document captures the complete design process behind the proposed hybrid YOLO architecture. It records the progression from literature analysis to module selection, architecture planning, integration strategy, and experimental design. Unlike the final thesis, its purpose is to serve as a practical engineering reference that documents the reasoning behind each architectural decision and provides a structured roadmap for implementation and evaluation.

